



Advanced Signal Processing

ES7, ES9

Fall 2024

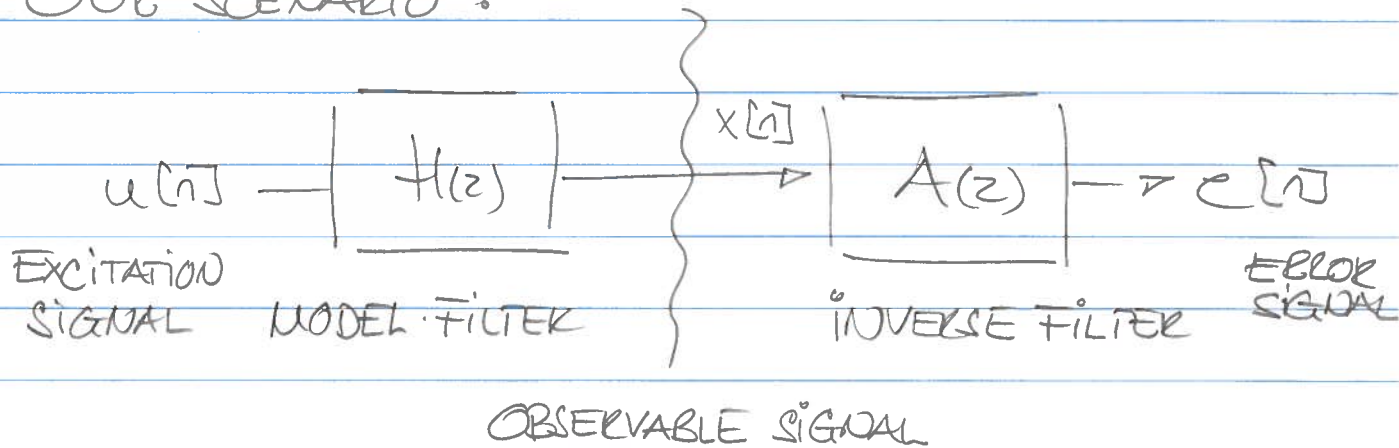
A Brief Intro to The Lattice Filter

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ADVANCED SIGNAL PROCESSING, ES-7

LINEAR PREDICTIVE CODING — LATTICE FILTERS.

OUR SCENARIO :



IN ORDER TO DETERMINE THE INVERSE FILTER $A(z)$ FOR A GIVEN MODEL ORDER p , WE WANT TO CONDUCT LMS OPTIMIZATION OF THE RESIDUAL SIGNAL $e[n]$.

WE SHOWED PREVIOUSLY THAT SUCH A PROCESS CAN BE CONSIDERED AS $A(z)$ BEING A FILTER WHICH "BEST POSSIBLE" IS WHITENING THE SIGNAL $x[n]$.

NO MATTER WHETHER WE LOOK AT THIS PROBLEM IN THE TIME- OR THE FREQUENCY DOMAIN, IN ALL BOILS DOWN TO A SET OF LINEAR EQUATIONS;

$$R \cdot a = -r$$

$$\underline{R} \cdot \underline{a} = -\underline{r}$$

WHERE $\underline{R}$ AND $\underline{r}$ HOLDS THE AUTO-CORRELATION COEFFICIENTS (AUTO-CORR METHOD), OR THE COVARIANCE COEFFICIENTS (COVARIANCE METHOD)

IN THE AUTO-CORRELATION CASE, $\underline{R}$ IS A SYMMETRIC TOEPLITZ MATRIX, AND THE SOLUTION I.E., $\underline{a}$, CAN BE FOUND USING THE LEVINSON-DURBIN ALGORITHM, (I. YULE-WALKER EQUATIONS);

$E_0 = r(0)$; * E is the Residual Variance, and r is the (normalized) autocorrelation, $r(i) = R(i)/R(0)$

* p is the model order

$$\text{for } m=1 \text{ to } p \quad k_m = \frac{r(m) - \sum_{k=1}^{m-1} a_k^{(m-1)} \cdot r(m-k)}{E_{m-1}};$$

$a_m^{(m)} = k_m$; * k is an auxiliary variable, which we will now denote the reflection coefficient.
* a is the m'th LPC coefficient for model order m

for $k=1$ to $m-1$

* this k is just a counter - don't

confuse it with the reflection coef.

$$a_k^{(m)} = a_k^{(m-1)} - k_m \cdot a_{m-k}^{(m-1)};$$

end

$$E_m = E_{m-1} (1 - k_m^2);$$

end

$$a_j = a_j^{(p)}; \quad j=1, 2, \dots, p$$

PC

LEVINSON-DURBIN ALGORITHM

(MAKHLOUL p. 566)
JACKSON p. 333
OLS p. 959

$$E_0 = r(0)^2;$$

$$\text{for } m=1 \text{ to } p \quad k_m = \frac{r(m) - \sum_{k=1}^{m-1} a_k^{(m-1)} \cdot r(m-k)}{E_{m-1}};$$

$$a_m^{(m)} = k_m;$$

for $k=1$ to $m-1$

$$a_k^{(m)} = a_k^{(m-1)} - k_m \cdot a_{m-k}^{(m-1)};$$

end

$$E_m = E_{m-1} (1 - k_m^2);$$

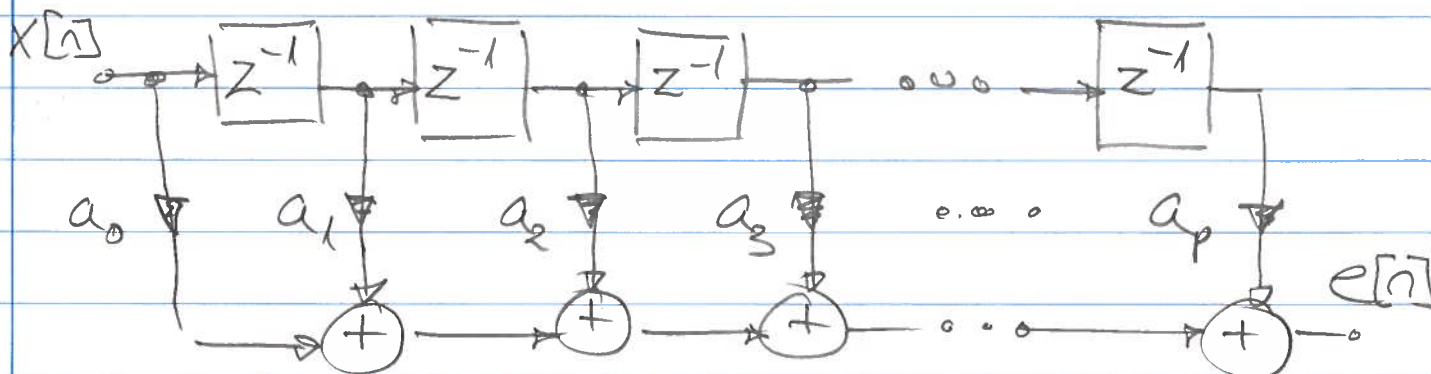
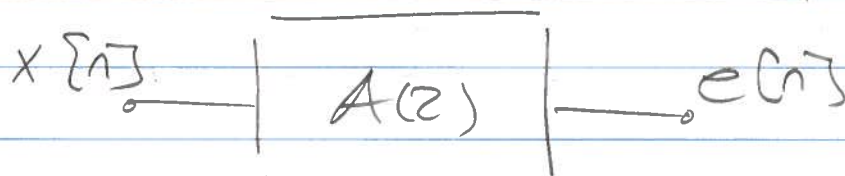
end

$$a_j = a_j^{(p)}; \quad j = 1, 2, \dots, p$$

THE LD-ALGORITHM IS AN ORDER-RECURSIVE UPDATE OF THE $A(z)$ -COEFFICIENTS, ALSO KNOWN AS THE LPC-COEFFICIENTS.

It means that the LPC coef. for model order m is a linear combination of the LPC coef. for model order $m-1$

WE WILL NOW USE THE LD-ALGORITHM TO DERIVE AN ALTERNATIVE STRUCTURE FOR $A(z)$, i.e., OTHER THAN THE TRANSVERSAL FILTER;



m^{th} ORDER PREDICTION ERROR SYSTEM;

$$X(z) \longrightarrow \boxed{A(z)^{(m)}} \longrightarrow E(z)^{(m)}$$

$$E(z)^{(m)} = X(z) \cdot A(z)^{(m)} = X(z) \cdot \left(1 + \sum_{k=1}^m a_k^{(m)} z^{-k} \right)$$

$z^{-1} \Downarrow$

$$e^{(m)}[n] = x[n] + \sum_{k=1}^m a_k^{(m)} x[n-k]$$

See also p.5 in the slides from lecture #1

THIS SIGNAL IS NOW DENOTED AS THE **FORWARD PREDICTION ERROR** FROM AN m^{th} ORDER LINEAR ONE-STEP PREDICTOR

... AND THE PREDICTION IS BASED ON A LINEAR COMBINATION OF m PREVIOUS VALUES OF $x[n]$.

WE NOW RE-WRITE THE POLYNOMIAL $A(z)$ USING THE STATEMENT (P.3)

$$a_k^{(m)} = a_k^{(m-1)} - k \cdot a_{m-k}^{(m-1)}$$

FROM THE LEVINSON-DORRIN ALGORITHM.

$$\begin{cases} A^{(m)}(z) = 1 + \sum_{k=1}^m a_k^{(m)} z^{-k} \\ a_k^{(m)} = a_k^{(m-1)} - k \cdot a_{m-k}^{(m-1)} \end{cases}$$

$$\Downarrow A^{(m)}(z) = 1 + \sum_{k=1}^m \left(a_k^{(m-1)} - k a_{m-k}^{(m-1)} \right) z^{-k}$$

$$\Downarrow A^{(m)}(z) = 1 + \sum_{k=1}^m a_k^{(m-1)} z^{-k} - k \cdot \sum_{k=1}^m a_{m-k}^{(m-1)} z^{-k}$$

SINCE THE LPC COEFF. ARE OF ORDER $m-1$, WE RECOGNIZE THIS TERM AS $A^{(m-1)}(z)$

BUT WHAT IS THIS SECOND TERM?

LET'S TRY TO WRITE OUT THE SUM

$$\sum_{k=1}^m a_{m-k}^{(m-1)} z^{-k}$$

$$= a_{m-1}^{(m-1)} z^{-1} + a_{m-2}^{(m-1)} z^{-2} + \dots + a_2^{(m-1)} z^{2-m} + a_1^{(m-1)} z^{1-m} + a_0^{(m-1)} z^{-m}$$

AND NOW IN REVERSE ORDER

$$= a_0^{(m-1)} z^{-m} + a_1^{(m-1)} z^{1-m} + a_2^{(m-1)} z^{2-m} + \dots + a_{m-2}^{(m-1)} z^{-2} + a_{m-1}^{(m-1)} z^{-1}$$

THE FIRST COEF IS ALWAYS EQU 1.

PK

6.

$$z^{-m} + a_1^{(m-1)} z^{1-m} + a_2^{(m-1)} z^{2-m} + \dots + a_{m-2}^{(m-1)} z^{-2} + a_{m-1}^{(m-1)} z^{-1}$$

$$= z^{-m} \left(1 + a_1^{(m-1)} z + a_2^{(m-1)} z^2 + \dots + a_{m-2}^{(m-1)} z^{m-2} + a_{m-1}^{(m-1)} z^{m-1} \right)$$

$$= z^{-m} \left(1 + \sum_{k=1}^{m-1} a_k^{(m-1)} z^k \right)$$

NOTE THAT THE SIGN
OF THE EXPONENT IS
POSITIVE

$$= z^{-m} A^{(m-1)}(z^{-1})$$

AND THEREFORE WE CAN NOW WRITE $A^{(m)}(z)$ AS;

$$\boxed{A^{(m)}(z) = A^{(m-1)}(z) - k_m z^{-m} A^{(m-1)}(z^{-1})} \quad \begin{matrix} \text{EQN 39} \\ \text{ORS p. 964} \end{matrix}$$

This is the "yellow equation" from p.5, but now with the right-hand side expressed as A-polynomials.

OK, WE NOW HAVE

$$\left\{ \begin{array}{l} \odot E^{(m)}(z) = A^{(m)}(z) \cdot X(z) \quad (\text{p.4}) \\ \odot A^{(m)}(z) = A^{(m-1)}(z) - k_m z^{-m} A^{(m-1)}(z^{-1}) \end{array} \right.$$

$$\Downarrow E^{(m)}(z) = \left(A^{(m-1)}(z) - k_m z^{-m} A^{(m-1)}(z^{-1}) \right) \cdot X(z)$$

$$\Downarrow E^{(m)}(z) = A^{(m-1)}(z) X(z) - k_m z^{-m} A^{(m-1)}(z^{-1}) X(z)$$

PK

7.

$$E^{(m)}(z) = A^{(m-1)}(z) \cdot X(z) - k_m \cdot z^{-m} A^{(m-1)}(z^{-1}) X(z)$$

$$X(z) \rightarrow \boxed{A^{(m-1)}(z)} \rightarrow E^{(m-1)}(z)$$

$(m-1)^{\text{st}}$ ORDER INVERSE FILTER

WHAT IS THIS TERM?

WE NOW DEFINE A NEW POLYNOMIAL;

$$\tilde{E}^{(m)}(z) \triangleq z^{-m} A^{(m)}(z^{-1}) X(z)$$

Note that the A-polynomial is order m.

LET'S HAVE A LOOK IN THE TIME-DOMAIN

$$\tilde{E}^{(m)}(z) = z^{-m} \left(1 + \sum_{k=1}^m a_k^{(m)} z^k \right) X(z)$$

$\Downarrow$

$$\tilde{E}^{(m)}(z) = X(z) z^{-m} + \sum_{k=1}^m a_k^{(m)} X(z) z^{k-m}$$

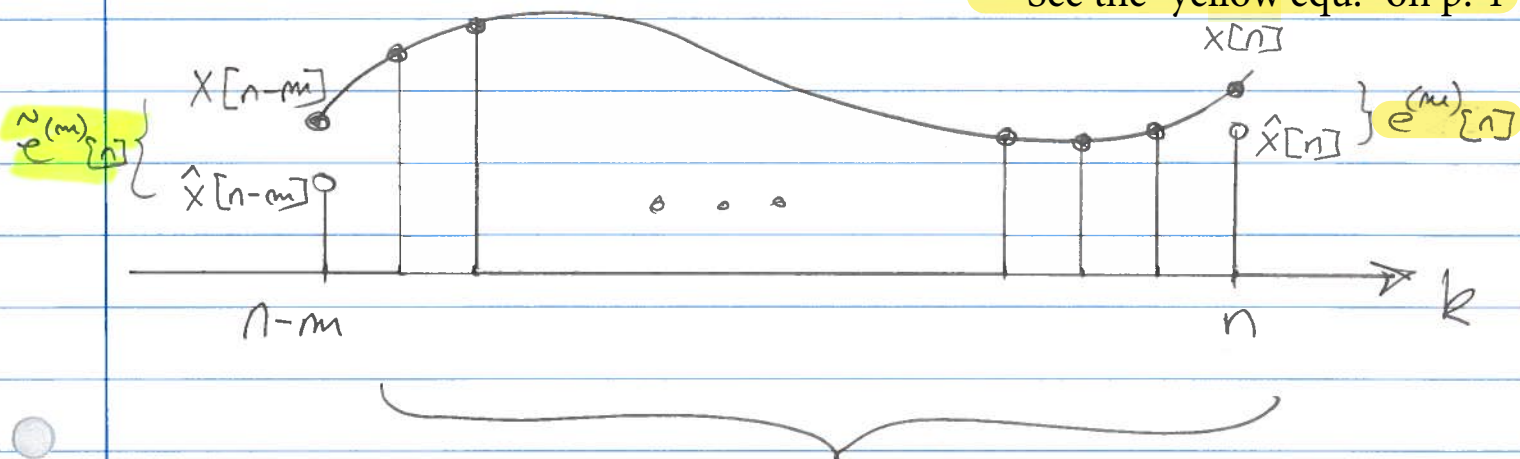
$z^{-1} \Downarrow$

$$\tilde{e}^{(m)}[n] = x[n-m] + \sum_{k=1}^m a_k^{(m)} x[n-m+k]$$

SO, OBVIOUSLY $\tilde{e}^{(m)}[n]$ IS A LINEAR COMBINATION OF DELAYED SAMPLES OF $x[n]$, BUT HOW SHOULD WE INTERPRETE IT...??

$$\tilde{e}^{(m)}[n] = X[n-m] + \sum_{k=1}^m a_k^{(m)} X[n-m+k]$$

See the "yellow equ." on p. 4



m samples

WHICH ARE USED TO PREDICT $X[n-m]$, i.e., $\hat{X}[n-m]$

SO, WHAT WE SEE HERE IS A PREDICTION OF THE OBSERVABLE SIGNAL WHICH POINTS IN A **BACKWARD DIRECTION**! BUT USING THE SAME LINEAR COMBINATION COEFFICIENTS a_k AS APPLIED FOR THE NORMAL FORWARD PREDICTION OF $x[n]$

THEREFORE, $\tilde{e}^{(m)}[n] = X[n-m] - \hat{X}[n-m]$ IS THE

BACKWARD PREDICTION ERROR

FOR AN m^{th} ORDER LINEAR ONE-STEP PREDICTOR

$$\hat{X}[n-m] = \sum_{k=1}^m a_k^{(m)} X[n-m+k]$$

Now, BACK TO p.7. (the top-most equation on p.7)

$$\begin{cases} E^{(m)}(z) = E^{(m-1)}(z) - k_m \underbrace{z^{-m} A^{(m-1)}(z^{-1}) X(z)}_{\text{This is the first "yellow equ.) on p.7}} \\ \tilde{E}^{(m)}(z) = z^{-m} A^{(m)}(z^{-1}) X(z) \end{cases}$$

WE WANT TO INSERT $\tilde{E}^{(m)}(z)$ INTO $E^{(m)}(z)$
BUT WE CANNOT DO THIS IMMEDIATELY. DUE
TO THE DIFFERENT ORDERS OF THE A POLYNOMIALS

HOWEVER;

$$\tilde{E}^{(m)}(z) = z^{-m} A^{(m)}(z^{-1}) X(z)$$

$$\Downarrow \quad \tilde{E}^{(m-1)}(z) = z^{-(m-1)} A^{(m-1)}(z^{-1}) X(z)$$

$$\Downarrow \quad \tilde{E}^{(m-1)}(z) = z \cdot z^{-m} A^{(m-1)}(z^{-1}) X(z)$$

$$\Downarrow \quad z^{-1} \tilde{E}^{(m-1)}(z) = z^{-m} A^{(m-1)}(z^{-1}) X(z)$$

AND THUS;

$$E^{(m)}(z) = E^{(m-1)}(z) - k_m z^{-1} \tilde{E}^{(m-1)}(z)$$

$z^{-1} \Downarrow$

$$\boxed{e^{(m)}[n] = e^{(m-1)}[n] - k_m \cdot e^{(m-1)}[n-1]}$$

FIRST PART OF A DIFFERENCE EQUATION.

Now, from p. 6 we have ;

$$A^{(m)}(z) = A^{(m-1)}(z) - k_m z^{-m} A^{(m-1)}(z^{-1}) \quad \text{EQU 89 OLS p. 964}$$

THIS EQUATION IS SUBSTITUTED INTO THE DEFINITION OF THE BACKWARD PREDICTION ERROR ON p. 7

$$\tilde{E}^{(m)}(z) \triangleq z^{-m} A^{(m)}(z^{-1}) X(z)$$

THE PROBLEM IS SIGN OF THE EXPONENTS ;
Z VERSUS z^{-1} .

$$A^{(m)}(z) = A^{(m-1)}(z) - k_m z^{-m} A^{(m-1)}(z^{-1})$$

⇓

$$A^{(m)}(z^{-1}) = A^{(m-1)}(z^{-1}) - k_m z^m A^{(m-1)}(z)$$

↑ NOW WE HAVE THE CORRECT SIGN.

$$\tilde{E}^{(m)}(z) = z^{-m} \left(A^{(m-1)}(z^{-1}) - k_m z^m A^{(m-1)}(z) \right) X(z)$$

⇓

$$\tilde{E}^{(m)}(z) = \left(z^{-m} A^{(m-1)}(z^{-1}) - k_m A^{(m-1)}(z) \right) X(z)$$

⇓

$$\tilde{E}^{(m)}(z) = \underbrace{z^{-m} A^{(m-1)}(z^{-1)} X(z)}_{\tilde{E}^{(m-1)}(z)} - k_m \underbrace{A^{(m-1)}(z) X(z)}_{E^{(m-1)}(z)}$$

⇓

$$\tilde{E}^{(m)}(z) = \tilde{E}^{(m-1)}(z) - k_m E^{(m-1)}(z)$$

(Last "yellow equation") see p. 9

see p. 4

$$\tilde{E}^{(m)}(z) = z^{-1} \tilde{E}^{(m-1)}(z) - k_m \cdot E^{(m-1)}(z)$$

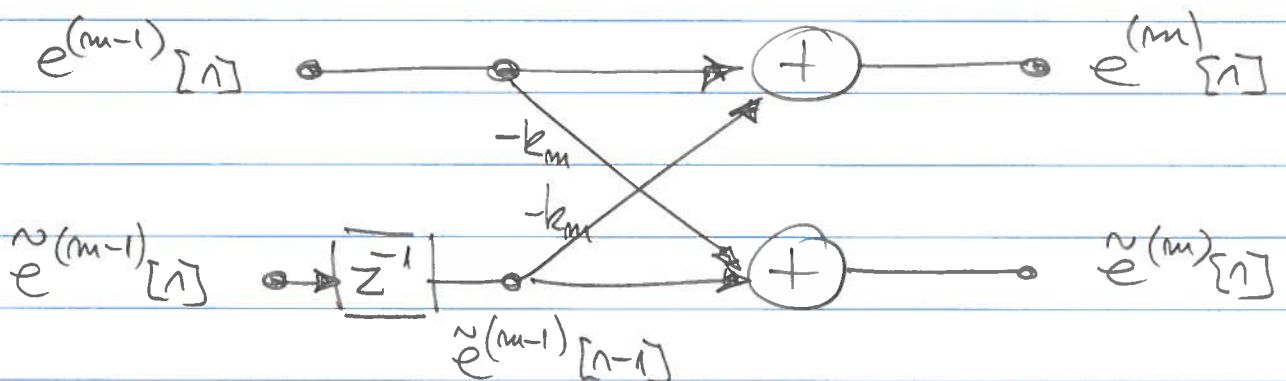
$$z^{-1} \Downarrow \left[\begin{array}{l} e^{(m)}[n] = e^{(m-1)}[n-1] - k_m e^{(m-1)}[n] \end{array} \right]$$

SECOND PART OF THE DIFFERENCE EQUATION.

AND NOW BOTH PARTS TOGETHER ;

$\begin{aligned} e^{(m)}[n] &= e^{(m-1)}[n] - k_m e^{(m-1)}[n-1] \\ \tilde{e}^{(m)}[n] &= \tilde{e}^{(m-1)}[n-1] - k_m e^{(m-1)}[n] \end{aligned}$	EQU. 95 AND 97 IN OBS
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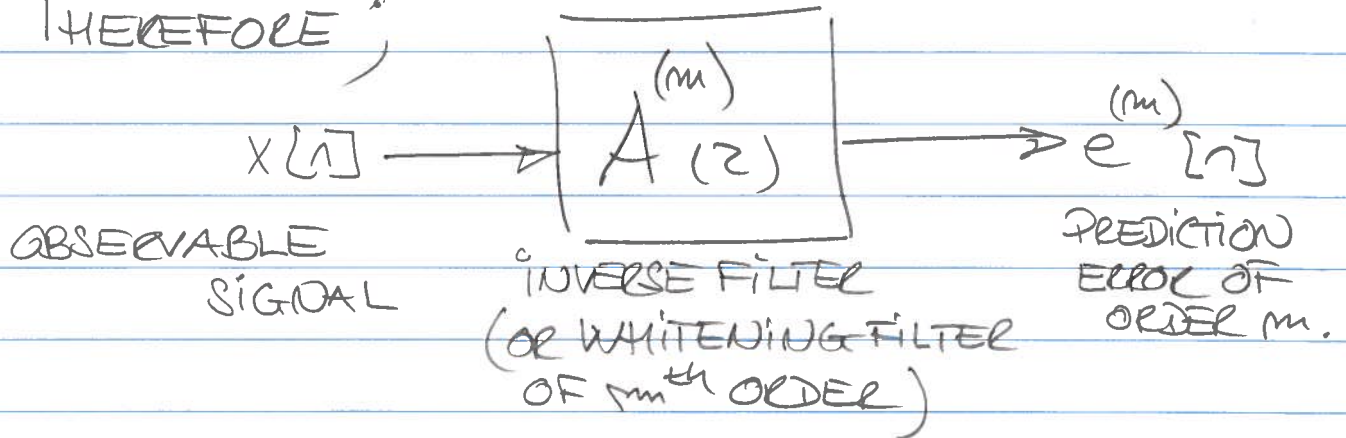
THIS SET OF DIFFERENCE EQUATIONS EXPRESSES THE m^{th} ORDER FORWARD AND BACKWARD PREDICTION ERRORS. GRAPHICAL DESCRIPTION ;



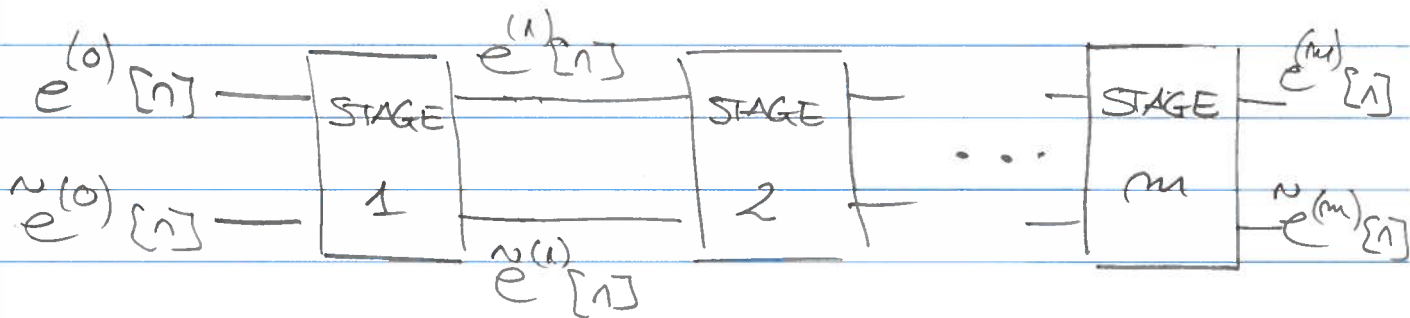
THIS GRAPH REPRESENTS ONE ITERATION OF THE LEVINSON-DURBIN ALG.

REMEMBER THAT THE LD ALG. IS AN ORDER-RECURSIVE UPDATE OF THE INVERSE FILTER $\hat{A}(z)$.

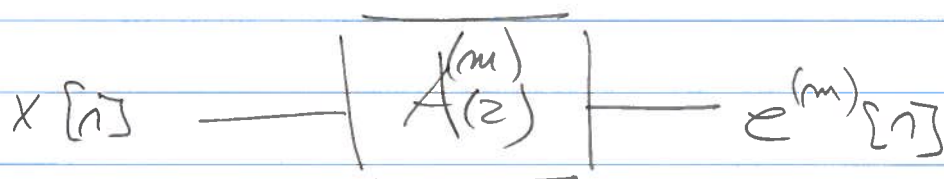
HEREFORE;



IN CONCLUSION, WE CAN NOW STATE THAT INSTEAD OF $A^{(m)}(z)$ BEING AN ORDINARY TAPPED-DELAY-LINE FIR-FILTER, IT COULD BE IMPLEMENTED AS A CASCADE OF m SECTIONS OF THE STRUCTURE ON p. 11 — THIS IS KNOWN AS THE LATTICE FILTER.

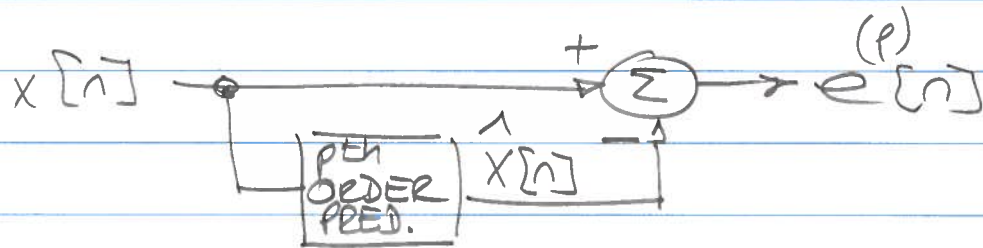


$e^{(m)}[n]$ IS THE OUTPUT OF THE FILTER — BUT WHAT IS THE INPUT ???



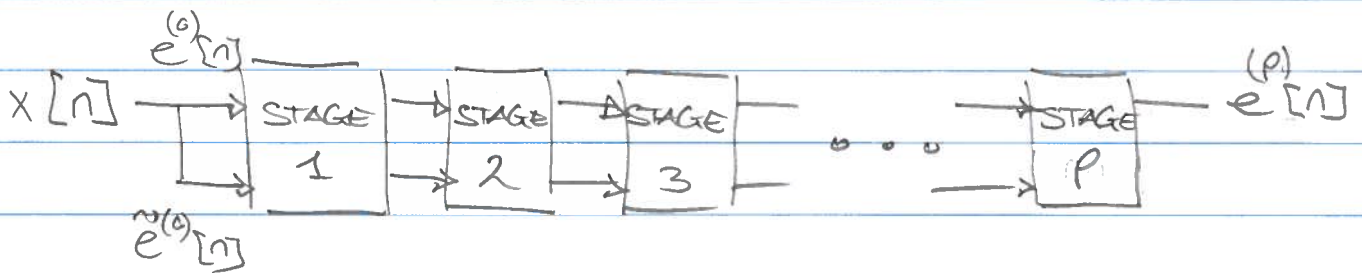
IF WE LOOK AT A p^{th} ORDER PREDICTOR THEN WE HAVE:

$$\hat{x}[n] = - \sum_{k=1}^p \alpha_k x[n-k]$$



IF NOW $p=0$ THEN WE CANNOT PREDICT, I.E., $\hat{x}[n]=0$ AND THUS $e^{(0)}[n] = x[n] - \hat{x}[n] = x[n]$

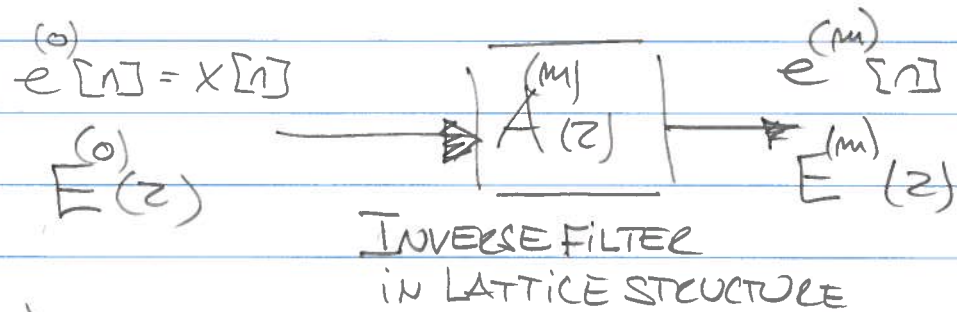
THE SAME ARGUMENT HOLDS FOR BACKWARD PREDICTION, AND THUS $\tilde{e}^{(0)}[n] = x[n]$



p^{th} ORDER FIR LATTICE FILTER

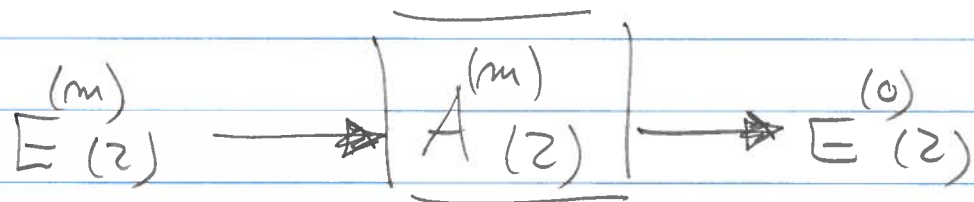
CAN WE ALSO DESIGN AN IIR LATTICE FILTER?

OUR USUAL SCENARIO ;



$$\frac{E^{(m)}(z)}{E^{(0)}(z)} = A^{(m)}(z) = 1 + \sum_{k=1}^m a_k^{(m)} z^{-k}$$

THE IDEA NOW IS TO MAINTAIN THE LATTICE STRUCTURE, BUT INTERCHANGE THE INPUT AND OUTPUT.



AND THUS $A^{(m)}(z) = \frac{E^{(0)}(z)}{E^{(m)}(z)}$.

THIS TRANSFER FUNCTION IS THE INVERSE OF WHAT WE HAD BEFORE, SO

$$A^{(m)}(z) = \frac{1}{1 + \sum_{k=1}^m a_k^{(m)} z^{-k}}$$

$$A(z) = \frac{E^{(0)}(z)}{E^{(m)}(z)} = \frac{1}{1 + \sum_{k=1}^m a_k^{(m)} z^{-k}}$$

IS NOW AR AND IDENTICAL TO THE AR MODEL FILTER $H(z)$ WITH $G=1$.

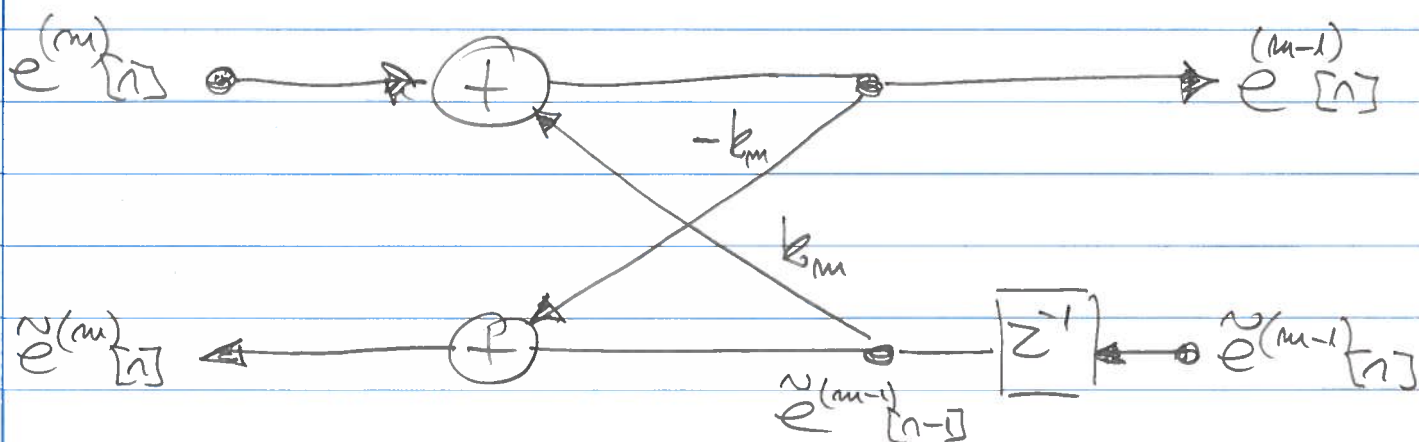
FROM THE LATTICE FIR-SCENARIO WE HAVE;

$$\begin{cases} e^{(m)}[n] = e^{(m-1)}[n] - k_m \tilde{e}^{(m-1)}[n-1] \\ \tilde{e}^{(m)}[n] = \tilde{e}^{(m-1)}[n-1] - k_m e^{(m-1)}[n] \end{cases}$$

NOW, ISOLATE $e^{(m-1)}[n]$ IN THE FIRST EQUATION

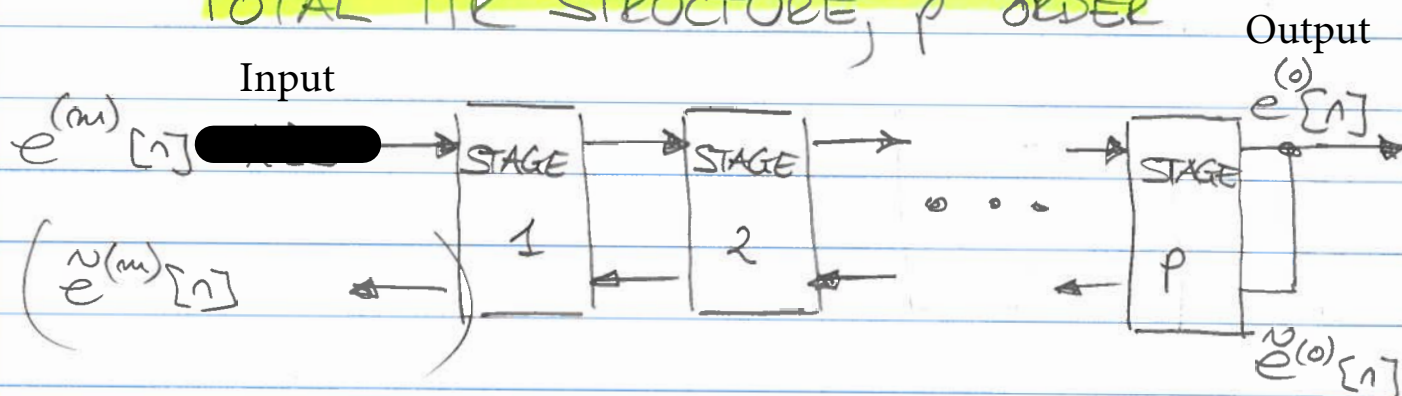
$$\begin{cases} e^{(m-1)}[n] = e^{(m)}[n] + k_m \tilde{e}^{(m-1)}[n-1] \\ \tilde{e}^{(m)}[n] = \tilde{e}^{(m-1)}[n-1] - k_m e^{(m-1)}[n] \end{cases}$$

GRAPHICAL REPRESENTATION



THIS IS A SINGLE STAGE OF AN FIR LATTICE FILTER

TOTAL IIR STRUCTURE, p^{th} ORDER



As for the FIR case, the 0th order forward and backward predictions are 0, and therefore $e[n]$ and $\tilde{e}[n]$ are identical and can thus be equalized to $x[n]$...

AS FOR THE FIR CASE, THE 0th FORWARD AND BACKWARD PREDICTION ERROR ARE 0.

DETERMINATION OF THE LATTICE FILTER (FIR) COEFFICIENTS, k_m

WITHOUT PROOF (OBS EQU 101, p. 967)

$$k_i = \frac{\sum_{n=-\infty}^{\infty} (e^{(i-1)}[n] \cdot \tilde{e}^{(i-1)}[n-1])}{\left\{ \sum_{n=-\infty}^{\infty} (e^{(i-1)}[n])^2 \cdot \sum_{n=-\infty}^{\infty} (\tilde{e}^{(i-1)}[n-1])^2 \right\}^{1/2}}$$

FORWARD RESIDUAL VARIANCE

BACKWARD RESIDUAL VARIANCE

NOTE THE i^{th} FILTER COEFFICIENT k_i IS CALCULATED BASED ON RESIDUAL SIGNALS FROM THE $(i-1)^{\text{st}}$ LATTICE STAGE!

TWO OBSERVATIONS :

① THE FORWARD AND BACKWARD RESIDUAL VARIANCES ARE IDENTICAL FOR ORDER m .

② DETERMINATION OF k_m IS BASED ON A FINITE SEGMENT OF THE OBSERVABLE SIGNAL.

$$k_m = - \frac{\sum_{n=0}^{N-1} (e^{(m-1)}[n] \cdot \tilde{e}^{(m-1)}[n-1])}{\sum_{n=0}^{N-1} (\tilde{e}^{(m-1)}[n])^2}$$